Lecture 36

Section 7.1

Diagonalization of Symmetric Matrices

Definition: A $n \times n$ matrix A is symmetric if $A = A^T$.

Examples:

$$\begin{bmatrix} 0 & -1 \\ -1 & 9 \end{bmatrix} \qquad \begin{bmatrix} 2 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 5 \end{bmatrix} \qquad \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$$

Goal: Diagonalize symmetric matrices nicely.

Theorem: If A is symmetric, then any two eigenvectors from different eigenspaces are orthogonal.

Proof: Suppose that $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors corresponding to distinct eigenvalues, namely λ_1 and λ_2, respectively. So, we have that $A\mathbf{v}_1 = \lambda_1\mathbf{v}_1$ and $A\mathbf{v}_2 = \lambda_2\mathbf{v}_2$. We want to show that $\mathbf{v}_1 \bullet \mathbf{v}_2 = 0$. Note that

$$\begin{aligned} \lambda_1(\mathbf{v}_1 \bullet \mathbf{v}_2) &= \lambda_1\mathbf{v}_1 \bullet \mathbf{v}_2 \\ &= (\lambda_1\mathbf{v}_1)^T\mathbf{v}_2 \\ &= (A\mathbf{v}_1)^T\mathbf{v}_2 \\ &= (\mathbf{v}_1^T A^T)\mathbf{v}_2 \\ &= \mathbf{v}_1^T(A\mathbf{v}_2) \\ &= \mathbf{v}_1^T(\lambda_2\mathbf{v}_2) \\ &= \lambda_2(\mathbf{v}_1^T\mathbf{v}_2) \\ &= \lambda_2(\mathbf{v}_1 \bullet \mathbf{v}_2). \end{aligned}$$

Hence, $\lambda_1(\mathbf{v}_1 \bullet \mathbf{v}_2) = \lambda_2(\mathbf{v}_1 \bullet \mathbf{v}_2)$, implying that

$$\lambda_1(\mathbf{v}_1 \bullet \mathbf{v}_2) - \lambda_2(\mathbf{v}_1 \bullet \mathbf{v}_2) = 0 \implies (\lambda_1 - \lambda_2)(\mathbf{v}_1 \bullet \mathbf{v}_2) = 0.$$

Because $\lambda_1 \neq \lambda_2$, then $\mathbf{v}_1 \bullet \mathbf{v}_2 = 0$. ■

Note: Recall that an orthogonal matrix A is an invertible square matrix such that $A^{-1} = A^T$. This is equivalent to A having orthonormal columns.

Definition: An $n \times n$ matrix A is orthogonally diagonalizable if $A = PDP^{-1} = PDP^T$ where P is an orthogonal matrix (i.e. $P^{-1} = P^T$) and D is a diagonal matrix.

Theorem: An $n \times n$ matrix A is orthogonally diagonalizable if and only if A is a symmetric matrix.

Example: Orthogonally diagonalize

$$A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}.$$

Solution: The characteristic equation of A is

$$\begin{aligned} 0 &= \det(A - \lambda I_2) \\ &= \begin{vmatrix} 7-\lambda & 3 \\ 3 & -1-\lambda \end{vmatrix} \\ &= -(7-\lambda)(1+\lambda) - 9 \\ &= -(7 + 7\lambda - \lambda - \lambda^2) - 9 \\ &= \lambda^2 - 6\lambda - 16 \\ &= (\lambda - 8)(\lambda + 2), \end{aligned}$$

and so the eigenvalues of A are $\lambda = -2, 8$.
Next, let's find the corresponding eigenvectors. Suppose $\lambda = -2$. Then,

$$A - (-2)I_2 = A + 2I_2 = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}.$$

Using row operations,

$$\begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix} \xrightarrow{r_2 \to r_2 - \frac{1}{3}r_1} \begin{bmatrix} 9 & 3 \\ 0 & 0 \end{bmatrix},$$

and so $9x_1 + 3x_2 = 0$ (i.e. $x_1 = -\frac{1}{3}x_2$), where x_2 is a free variable. Therefore,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3}x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -\frac{1}{3} \\ 1 \end{bmatrix}.$$

So, $\left\{ \begin{bmatrix} -\frac{1}{3} \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace corresponding to $\lambda = -2$.
Suppose $\lambda = 8$. Then,

$$A - 8I_2 = \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix}.$$

Using row operations,

$$\begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix} \xrightarrow{r_2 \to r_2 + 3r_1} \begin{bmatrix} -1 & 3 \\ 0 & 0 \end{bmatrix},$$

and so $-x_1 + 3x_2 = 0$ (i.e. $x_1 = 3x_2$), where x_2 is a free variable. Therefore,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

So, $\left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace corresponding to $\lambda = 8$.

By our first theorem from today's lecture, $\mathbf{v}_1 = \begin{bmatrix} -\frac{1}{3} \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ are orthogonal. We see that

$$||\mathbf{v}_1|| = \sqrt{\mathbf{v}_1 \bullet \mathbf{v}_1} = \sqrt{\frac{1}{9} + 1} = \sqrt{\frac{10}{9}} = \frac{\sqrt{10}}{3} \implies \frac{\mathbf{v}_1}{||\mathbf{v}_1||} = \begin{bmatrix} -\frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix},$$

$$||\mathbf{v}_2|| = \sqrt{\mathbf{v}_2 \bullet \mathbf{v}_2} = \sqrt{9+1} = \sqrt{10} \implies \frac{\mathbf{v}_2}{||\mathbf{v}_2||} = \begin{bmatrix} \frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix}.$$

So, we let

$$P = \begin{bmatrix} -\frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 \\ 0 & 8 \end{bmatrix},$$

where $A = PDP^T$.

Note: Eigenvectors corresponding to different eigenvalues are orthogonal. If you have an eigenvalue of multiplicity 2 or higher, your vectors generating the eigenspace corresponding to that eigenvalue may not be orthogonal. You might have to use Gram-Schmidt to make the eigenbasis orthogonal.

Example: Suppose

$$A = \begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix}.$$

The characteristic polynomial of A is

$$-\lambda^3 + 18\lambda^2 - 81\lambda.$$

The eigenvalues corresponding to A are $\lambda = 0, 9, 9$.

The eigenbasis corresponding to $\lambda = 0$ is $\left\{ \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \right\}$. The eigenbasis corresponding to $\lambda = 9$ is $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \right\}$. Notice that

$$\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \bullet \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \neq 0.$$

We need to use Gram-Schmidt to make the eigenbasis corresponding to $\lambda = 9$ an orthogonal set. So,

$$\begin{aligned}
\mathbf{u}_1 &= \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \\
\mathbf{u}_2 &= \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} - \frac{\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \bullet \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \bullet \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 2 \end{bmatrix}.
\end{aligned}$$

Therefore,

$$\left\{ \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 2 \end{bmatrix} \right\}$$

is a set of orthogonal vectors.

The Spectral Theorem: Let A be an $n \times n$ symmetric matrix. Then,

1. A has n real eigenvalues (counting multiplicities),
2. the dimension of the eigenspace for each eigenvalue λ is equal to the multiplicity of λ as a root of the characteristic equation,
3. the eigenspaces are mutually orthogonal,
4. A is orthogonally diagonalizable.

Spectral Decomposition

Suppose that $A = PDP^{-1}$ such that the columns of P are orthonormal vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ of A and the corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$ are in the diagonal matrix D. Then, we

have that

$$
\begin{aligned}
A &= PDP^{-1} \\
&= PDP^T \\
&= \begin{bmatrix} \mathbf{u}_1 \cdots \mathbf{u}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} \\
&= \begin{bmatrix} \lambda_1 \mathbf{u}_1 \cdots \lambda_n \mathbf{u}_n \end{bmatrix} \begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} \\
&= \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \cdots + \lambda_n \mathbf{u}_n \mathbf{u}_n^T.
\end{aligned}
$$

We call this the spectral decomposition of A.

Example: Suppose

$$
A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}.
$$

From our first example today, we have that $A = PDP^T$, where

$$
P = \begin{bmatrix} -\frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 \\ 0 & 8 \end{bmatrix}.
$$

Let

$$
\mathbf{u}_1 = \begin{bmatrix} -\frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix}, \qquad \mathbf{u}_2 = \begin{bmatrix} \frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix}.
$$

Then, the spectral decomposition of A is

$$
\begin{aligned}
A &= -2 \begin{bmatrix} -\frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{bmatrix} + 8 \begin{bmatrix} \frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} \\
&= -2 \begin{bmatrix} \frac{1}{10} & -\frac{3}{10} \\ -\frac{3}{10} & \frac{9}{10} \end{bmatrix} + 8 \begin{bmatrix} \frac{9}{10} & \frac{3}{10} \\ \frac{3}{10} & \frac{1}{10} \end{bmatrix}.
\end{aligned}
$$